

DATABASE MANAGEMENT SYSTEMS (CS301)

Normalization Guidelines (1NF to BCNF)

Learning Objectives

Understand normalization, functional dependencies, normal forms, anomalies, and database design improvements.

Introduction to Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

Need for Normalization

- Reduce data redundancy
- Eliminate anomalies
- Improve consistency
- Simplify database maintenance

Data Anomalies

Update Anomaly: Same data must be updated in multiple places.

Insert Anomaly: Cannot insert data without other related data.

Delete Anomaly: Deleting a record causes loss of important information.

First Normal Form (1NF)

A relation is in 1NF if all attributes contain atomic values and there are no repeating groups.

Second Normal Form (2NF)

A relation is in 2NF if it is in 1NF and has no partial dependency on a composite key.

Third Normal Form (3NF)

A relation is in 3NF if it is in 2NF and has no transitive dependency.

Boyce-Codd Normal Form (BCNF)

A relation is in BCNF if for every functional dependency $X \rightarrow Y$, X is a super key.

Example of Normalization

Unnormalized Table: Student(StudentID, Name, Course1, Course2)

1NF: Separate repeating course values.

2NF: Remove partial dependencies.

3NF: Remove transitive dependencies.

BCNF: Ensure every determinant is a candidate key.

Comparison of Normal Forms

1NF → Atomic values only

2NF → No partial dependency

3NF → No transitive dependency

BCNF → Stronger version of 3NF

Advantages of Normalization

- Reduces redundancy
- Improves consistency
- Enhances integrity
- Easier maintenance

Applications

- Banking Databases
- University Management Systems
- Hospital Information Systems
- E-Commerce Platforms

Exam Important Questions

1. Define normalization.
2. Explain anomalies with examples.
3. Differentiate 1NF, 2NF, and 3NF.
4. Explain BCNF.
5. Normalize a given relation up to BCNF.

Summary

Normalization is a key database design technique used to minimize redundancy and improve consistency. Understanding 1NF, 2NF, 3NF, and BCNF is essential for designing efficient relational databases.